

Today

Runge-Kutta

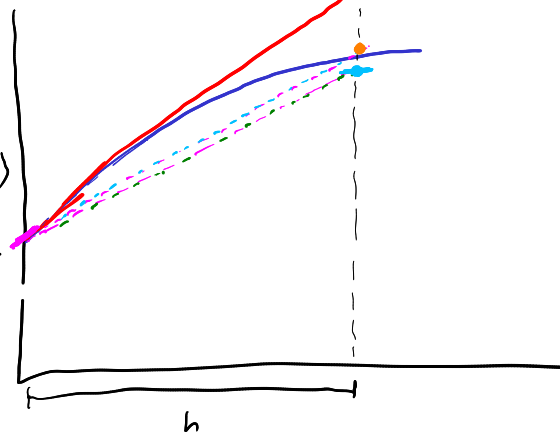
Heun's algorithm

Step i:

predict: $x_{i+1}^0 \leftarrow x_i^m + f(t_i, x_i)h$

Repeat until $\left| \frac{x_{i+1}^{j+1} - x_{i+1}^j}{x_{i+1}^j} \right| < \epsilon_s$: (n iterations over j)

correct: $x_{i+1}^j \leftarrow x_i^m + h \frac{f(t_i, x_i^m) + f(t_{i+1}, x_{i+1}^{j-1})}{2}$
 $j \leftarrow j+1$



Local: $O(h^3)$ Global: $O(h^2)$

Runge Kutta

Classical RK4 "4th order Runge Kutta"

update

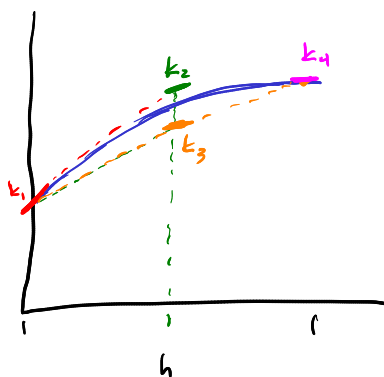
$x_{i+1} \leftarrow x_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)h$

$k_1 = f(t_i, x_i)$

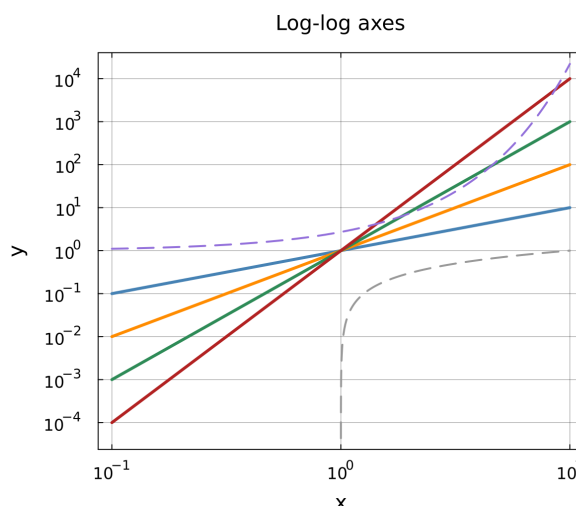
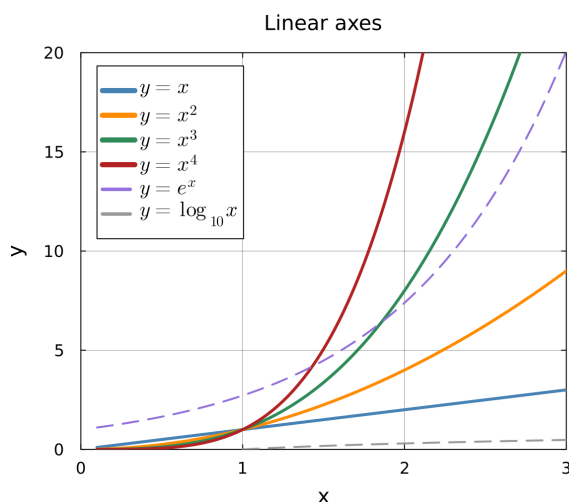
$k_2 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_1h)$

$k_3 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_2h)$

$k_4 = f(t_i + h, x_i + k_3h)$



Local: $O(h^5)$ Global: $O(h^4)$



coordinates on log-log

$x' = \log(x)$ $y' = \log(y)$

$y' = \log(x^k) = k \log(x) = kx'$

$y' = kx'$

$y = x^k$

General Runge-Kutta

$$x_{i+1} \leftarrow x_i + \phi h$$

$$\phi = a_1 k_1 + a_2 k_2 + \dots + a_n k_n$$

$$k_1 = f(t_i, x_i)$$

$$k_2 = f(t_i + p_1 h, x_i + q_{11} k_1 h)$$

$$k_3 = f(t_i + p_2 h, x_i + q_{21} k_1 h + q_{22} k_2 h)$$

⋮

$$k_n = f(t_i + p_{n-1} h, x_i + q_{n-1,1} k_1 h + \dots + q_{n-1,n-1} k_{n-1} h)$$

How to find a's, p's, q's?

Set equal to order-n Taylor Series

Example: Second Order

$$x_{i+1} \leftarrow x_i + (a_1 k_1 + a_2 k_2) h$$

$$f_i \equiv f(t_i, x_i) = k_1$$

$$k_2 = f(t_i + p h, x_i + q k_1 h)$$

$$x_{i+1} = x_i + a_1 f_i h + a_2 f(t_i + p h, x_i + q f_i h) h$$

$$= x_i + a_1 f_i h + a_2 \left(f_i + \frac{\partial f}{\partial t} \Big|_i p h + \frac{\partial f}{\partial x} \Big|_i q f_i h \right) h$$

Taylor Series

$$x_{i+1} \approx x_i + \frac{dx}{dt} \Big|_i h + \frac{d^2 x}{dt^2} \Big|_i \frac{h^2}{2} + \dots$$

$$= x_i + f_i h + \left(\frac{\partial f}{\partial t} \Big|_i + \frac{\partial f}{\partial x} \Big|_i f_i \right) \frac{h^2}{2}$$

equal to each other

$$a_1 + a_2 = 1$$

$$p h^2 a_2 = \frac{h^2}{2}$$

$$p a_2 = \frac{1}{2}$$

$$a_2 q f_i h^2 = f_i \frac{h^2}{2}$$

$$a_2 q = \frac{1}{2}$$

$$a_1 + a_2 = 1$$

$$a_2 p = \frac{1}{2}$$

$$a_2 q = \frac{1}{2}$$

3 equations
4 unknowns

under-determined

$$\textcircled{1} \quad a_2 = \frac{1}{2} \Rightarrow a_1 = \frac{1}{2}, \quad p = q = 1$$

Heun's with no iter

$$\textcircled{2} \quad a_2 = 1 \Rightarrow a_1 = 0, \quad p = q = \frac{1}{2}$$

Midpoint

$$\textcircled{3} \quad a_2 = \frac{3}{4} \Rightarrow a_1 = \frac{1}{4}, \quad p = q = \frac{2}{3}$$

Ralston's

First order

$$x_{i+1} \leftarrow x_i + k_1 h$$

Euler's

second order truncation

⇒ second order RK method

chain rule

$$\begin{aligned} \frac{d^2 x}{dt^2} \Big|_i &= \frac{df}{dt} \Big|_i = \frac{\partial f}{\partial t} \Big|_i + \frac{\partial f}{\partial x} \Big|_i \frac{dx}{dt} \Big|_i \\ &= \frac{\partial f}{\partial t} \Big|_i + \frac{\partial f}{\partial x} \Big|_i f_i \end{aligned}$$